

Spectral Rigidity and Crystallization $\mathrm{PSL}(2,7)$ on Seven Points: **in the Coset Geometry of $\mathrm{PSL}(2,7)$** **A Spectral Record of the** **Gelfand Pair $(\mathrm{PSL}(2,7), S_4)$**

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Abstract

Let $G = \mathrm{PSL}(2,7) \cong \mathrm{GL}(3,2)$ be the unique simple group of order 168. We study the coset geometry arising from the maximal subgroup $S_4 < G$ of index 7, and establish record four main results. First, among the maximal subgroups of G , the subgroup S_4 — in either of its two conjugacy classes, the point- and line-stabilizers interchanged by the outer automorphism — is characterized by the property that the six-dimensional irreducible representation ρ_6 contains its trivial character exactly once (Theorem 2.3). Second, we provide explicit $\mathrm{GL}(3,2)$ generators for $S_4 = \mathrm{Stab}(P_1)$ in the Fano plane $\mathrm{PG}(2,2)$, with all presentation relations verified by direct matrix computation (Lemma 2.5). Third, the first non-trivial eigenvalue of the coset Laplacian on $L^2(G/S_4)$ is $\lambda_1 = 2/3$, and this spectral gap is stable under any G -invariant perturbation preserving the generating set, by virtue of the multiplicity-free decomposition $L^2(G/S_4) \cong \rho_1 \oplus \rho_6$ and Schur's lemma (Lemma 2.9). Fourth, the random-walk Cheeger constant of the 7-vertex coset graph is $h = 8/21$, and the discrete Cheeger inequality $h^2/2 \leq \lambda_1 \leq 2h$ holds, with neither bound attained (Theorem 3.3). As a separate structural result, we establish record via Smith's theorem that $\mathrm{PSL}(2,7)$ cannot act freely on any sphere, so that $S^7/\mathrm{PSL}(2,7)$ is an orbifold, not a manifold. Explicit generators for the flag stabilizer $D_4 = \mathrm{Stab}(P_1, L_{124})$ are supplied in an appendix. The statements of Theorem 2.3 and Lemmas 2.8–2.9 are instances of classical facts: G acts 2-transitively on the seven cosets of S_4 , so (G, S_4) is a Gelfand pair, and the distance-transitive representations of the groups G with $\mathrm{PSL}_2(q) \triangleleft G \leq \mathrm{P}\Gamma\mathrm{L}_2(q)$ are classified [15]; this note records the case $q = 7$ in spectral language.

These results are of independent mathematical interest; their potential physical interpretation requires additional theoretical development and is not assumed within the scope of this work.

Keywords: $\mathrm{PSL}(2,7)$; finite simple groups; Fano plane; coset geometry; spectral gap; Cheeger constant

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1. Introduction

The group $G = \mathrm{PSL}(2,7)$, the projective special linear group over the field of seven elements, is the unique simple group of order $168 = 2^3 \cdot 3 \cdot 7$. Through the isomorphism $\mathrm{PSL}(2,7) \cong \mathrm{GL}(3,2)$ [4], it also acts faithfully as the automorphism group of the Fano plane $\mathrm{PG}(2, \mathbb{Z}_2)$, the projective plane over $\mathbb{Z}_2$ whose seven points and seven lines admit a transitive, indeed 2-transitive, collineation group of order 168. This dual identity — as a group of Möbius transformations on $\mathbb{P}^1(\mathbb{Z}_7)$ and as a group of projective symmetries of the smallest projective plane — makes $\mathrm{PSL}(2,7)$ a natural meeting point of classical number theory, finite combinatorics, and spectral geometry.

The maximal subgroup structure of G is completely determined by Dickson's classification of the maximal subgroups of $\mathrm{PSL}(2,q)$ for prime powers q [6, 7]. For $q = 7$, there are exactly three conjugacy classes of maximal subgroups, comprising two classes of S_4 and one of the Borel subgroup:

$$S_4 = \mathrm{Stab}(P) \text{ (point-stabilizer, order 24, index 7), } S_4 = \mathrm{Stab}(\ell) \text{ (line-stabilizer, order 24, index 7), } \\ \mathbb{Z}_7 \rtimes \mathbb{Z}_3 \text{ (order 21, index 8).}$$

Among these, the symmetric group S_4 occupies a distinguished spectral position that we call *spectral crystallization*: among the maximal subgroups of G , the property that the 6-dimensional irreducible representation ρ_6 of G contains the trivial representation of H with multiplicity exactly one picks out precisely the two conjugacy classes of S_4 , the point- and line-stabilizers in $\mathrm{PG}(2,2)$, and excludes the Borel subgroup $\mathbb{Z}_7 \rtimes \mathbb{Z}_3$.

The first goal of this paper is to establish this characterization precisely, by direct Frobenius reciprocity calculations for each maximal subgroup class (Theorem 2.3), and to record that the two S_4 classes are interchanged by the outer automorphism of G — the algebraic form of Fano-plane point-line duality. The second goal is to realize the characterization concretely within the Fano plane geometry, providing explicit $\mathrm{GL}(3,2)$ generator matrices for S_4 as the point stabilizer $\mathrm{Stab}(P_1)$, with all presentation and order relations verified by direct matrix computation over $\mathbb{Z}_2$ (Lemma 2.5).

The spectral content of the coset space G/S_4 is simple: by Frobenius reciprocity, $L^2(G/S_4) \cong \rho_1 \oplus \rho_6$, with all other irreducible representations of G absent. The 7-dimensional coset Laplacian therefore has exactly one non-trivial eigenvalue $\lambda_1 = 2/3$, computed from the character value $\chi_{\rho_6}(2A) = 2$ via Schur's lemma. Lemma 2.9 shows this gap is rigid: any G -invariant perturbation of the Laplacian preserving the generating set cannot alter the eigenvalue on the ρ_6 -isotypic component.

In Section 3 we analyze the 7-vertex coset graph. The 2-transitive action of G on $\mathrm{PG}(2,2)$ forces the graph to be the complete multigraph $2 \cdot K_7$ (every pair of vertices connected by exactly two edges). We compute the random-walk Cheeger constant $h = 8/21$ and verify the discrete Cheeger inequality $h^2/2 \leq \lambda_1 \leq 2h$ [3], noting that neither bound is attained.

In Section 4 we maintain a careful separation among three spectral quantities that arise naturally in this geometry: the coset Laplacian eigenvalue $\lambda_1 = 2/3$, the inter-representation gap $\delta = 10/21$ in the full $\mathrm{PSL}(2,7)$ spectrum, and the Cheeger constant $h = 8/21$. These are conceptually distinct objects, and we flag the numerical coincidence $h_2 = \delta = 10/21$ (where h_2 denotes the Cheeger ratio at 2-vertex cuts) as a structural open question. We also establish, as an independent result via Smith's theorem,

that $\mathrm{PSL}(2,7)$ cannot act freely on any sphere, so $S^7/\mathrm{PSL}(2,7)$ is an orbifold. The orbifold index theory of this space is identified as an open problem requiring methods beyond the scope of this paper.

Prior art. The spectral statements of Section 2 are instances of classical facts, recorded here for $q = 7$. The group G acts 2-transitively on the seven cosets of S_4 , so the permutation character is $1 + \chi_{\rho_6}$ (Lemma 2.8) and (G, S_4) is a Gelfand pair; the rigidity of Lemma 2.9 is Schur’s lemma for this rank-2 action, and the eigenvalue $2/3$ is read off from the central character of the class $2A$ on ρ_6 . Multiplicity-free permutation representations of the finite linear groups were studied by Inglis, Liebeck and Saxl [16]. Praeger, Saxl and Yokoyama [18] reduced the classification of primitive distance-transitive graphs to the almost simple and affine cases, and Faradžev and Ivanov [15] classified the distance-transitive representations of the groups G with $\mathrm{PSL}_2(q) \triangleleft G \leq \mathrm{P}\Gamma\mathrm{L}_2(q)$. The terms *spectral rigidity* and *spectral crystallization* used in this note are descriptive names for these facts; no further theorem is attached to them, and no novelty is claimed for these statements.

Conventions. All groups are finite. Matrices are written over $\mathbb{Z}_2$ unless stated otherwise, with arithmetic modulo 2. The inner product of class functions χ, ψ on a finite group H is $\langle \chi, \psi \rangle_H = (1/|H|) \sum_{h \in H} \chi(h)\psi(h)$. Conjugacy class labels (1A, 2A, etc.) follow the ATLAS of Finite Groups [5]. The character table of $\mathrm{PSL}(2,7)$ appears in [5] and is recalled in Table 1 below.

2. Main Results

2.1 Subgroup Classification

We recall the relevant structure of $G = \mathrm{PSL}(2,7)$.

Theorem 2.1 (Dickson [6]; Huppert [7], Satz II.8.27; Wilson [12], §3.3.5). *The maximal subgroups of $\mathrm{PSL}(2,7)$, up to conjugacy, are exactly:*

Label	Isomorphism type	Order	Index
M_1	$S_4 = \mathrm{Stab}(P)$	24	7
M_2	$Z_7 \rtimes Z_3$	21	8
M_3	$S_4 = \mathrm{Stab}(\ell)$	24	7

Remark 2.2. The two classes of S_4 are the point-stabilizers $\mathrm{Stab}(P)$ and the line-stabilizers $\mathrm{Stab}(\ell)$ in $\mathrm{PG}(2,2)$. They are abstractly isomorphic but not conjugate in G : the outer automorphism of $\mathrm{PSL}(2,7) \cong \mathrm{GL}(3,2)$, realized algebraically as the inverse-transpose map $g \mapsto (g^{-1})^T$ — the Fano-plane point-line duality — interchanges them, while fixing the Borel class $Z_7 \rtimes Z_3$ (the Singer-cycle normalizer). The dihedral group D_4 of order 8 is the Sylow 2-subgroup of G and is not maximal: since $7 \equiv -1 \pmod{8}$, G contains S_4 , and D_4 is the Sylow 2-subgroup of each S_4 , so $D_4 \subsetneq S_4 \subsetneq G$. It appears in this work only as the flag stabilizer $\mathrm{Stab}(P_1, L_{124})$ of index 21 (Appendix A), not as a maximal subgroup. The irreducible complex representations of G have dimensions 1, 3, 3, 6, 7, 8, which we denote $\rho_1, \rho_3, \rho_3', \rho_6, \rho_7, \rho_8$ respectively. The dimension identity $1 + 9 + 9 + 36 + 49 + 64 = 168$ confirms six conjugacy classes.

Table 1: Conjugacy class sizes and ρ_6 character values for $\mathrm{PSL}(2,7)$ [5].

Class	1A	2A	3A	4A	7A/7B
Class size	1	21	56	42	24 each
χ_{ρ_6}	6	2	0	0	-1

Remark 2.2b (Fusion of conjugacy classes). For any prime $p \geq 5$, the group $\text{PSL}(2,p)$ has a unique conjugacy class of involutions. In $G = \text{PSL}(2,7)$, there is therefore a unique class of elements of order 2, labeled 2A, of size 21. Consequently, every element of order 2 in any subgroup $H < G$ fuses to class 2A under the inclusion $H \hookrightarrow G$ [1]. Similarly, G has a unique class 3A of elements of order 3 and a unique class 4A of elements of order 4, so elements of those orders in subgroups fuse accordingly.

2.2 The S_4 Characterization Theorem

For a subgroup $H < G$ and an irreducible representation (ρ, V) of G , define the *trivial multiplicity*

$$n_1(H; \rho) = \langle \chi_\rho|_H, 1_H \rangle_H = (1/|H|) \sum_{h \in H} \chi_\rho(h).$$

By Frobenius reciprocity [10, Theorem 7.2], $n_1(H; \rho)$ equals the multiplicity of ρ in $\text{Ind}^H 1_H \cong L^2(G/H)$.

Theorem 2.3 (S_4 Characterization). *Among the maximal subgroups of $G = \text{PSL}(2,7)$, the subgroup S_4 — in each of its two conjugacy classes (the point- and line-stabilizers, interchanged by the outer automorphism) — satisfies $n_1(H; \rho_6) = 1$, and these are the only maximal subgroups with this property; the Borel subgroup $Z_7 \rtimes Z_3$ has $n_1(H; \rho_6) = 0$. Moreover, the two classes of S_4 are exactly the maximal subgroups of G whose order is divisible by 3 and which contain elements of order 4.*

Proof. We check the maximal subgroup classes from Theorem 2.1, using the character values in Table 1 and the fusion of Remark 2.2b. Since $n_1(H; \rho)$ depends only on the conjugacy classes of G met by H (through the character χ_ρ), it is constant across isomorphic subgroups with identical fusion; in particular the two S_4 classes share the same value.

Case 1: $H = M_1 = S_4$ (point stabilizer), $|H| = 24$. The group S_4 has element orders: 1 identity, 6 transpositions, 3 double transpositions (2-cycles paired), 8 elements of order 3, and 6 elements of order 4. All 9 elements of order 2 fuse to 2A; the 8 elements of order 3 fuse to 3A; the 6 elements of order 4 fuse to 4A. Therefore:

$$n_1(S_4; \rho_6) = (1/24)[1 \cdot 6 + 9 \cdot 2 + 8 \cdot 0 + 6 \cdot 0] = 24/24 = 1. \quad \checkmark$$

Case 2: $H = M_2 = Z_7 \rtimes Z_3$, $|H| = 21$. The group $Z_7 \rtimes Z_3$ has element orders: 1 identity, 14 elements of order 3 (fusing to 3A), and 6 elements of order 7 (fusing to class 7A or 7B). There are no elements of order 2 or 4. Therefore:

$$n_1(M_2; \rho_6) = (1/21)[1 \cdot 6 + 14 \cdot 0 + 6 \cdot (-1)] = 0/21 = 0.$$

Case 3: $H = M_3 = S_4$ (line stabilizer), $|H| = 24$. The line-stabilizer is abstractly isomorphic to the point-stabilizer of Case 1, with the same element-order distribution and the same fusion into the classes 2A, 3A, 4A of G . The outer automorphism $g \mapsto (g^{-1})^T$ carries M_1 to M_3 and fixes the character χ_{ρ_6} , so:

$$n_1(M_3; \rho_6) = n_1(M_1; \rho_6) = 1. \quad \checkmark$$

Thus $n_1(H; \rho_6) = 1$ for both S_4 classes (M_1 and M_3), while $n_1(M_2; \rho_6) = 0$ for the Borel subgroup. Among the maximal subgroups of G , the value $n_1(\cdot; \rho_6) = 1$ therefore characterizes exactly the S_4 isomorphism type. The divisibility statement follows similarly: both S_4 classes have order $24 = 2^3 \cdot 3$ and contain elements of order 4, whereas the Borel M_2 has order $21 = 3 \cdot 7$ and no element of order 4.

□

Remark 2.3b. The condition $n_1(\cdot; \rho_6) = 1$ is genuinely restrictive even among small subgroups. The (non-maximal) Sylow 2-subgroup $D_4 < S_4$, which appears as the flag stabilizer of Appendix A, has 1 identity, 5 involutions (fusing to 2A), and 2 elements of order 4 (fusing to 4A), giving $n_1(D_4; \rho_6) = (1/8)[1 \cdot 6 + 5 \cdot 2 + 2 \cdot 0] = 16/8 = 2$. Thus the value 1 is not automatic for subgroups that contain both involutions and order-4 elements; it is specific to the maximal S_4 classes.

2.3 Constructive $\mathrm{GL}(3,2)$ Generators for S_4

We realize the S_4 characterization concretely via the Fano plane geometry.

Definition 2.4 (Fano plane). The Fano plane $\mathrm{PG}(2,2)$ consists of the 7 nonzero vectors in $\mathbb{Z}_2^3$, labeled:

$$P_1 = (1,0,0), P_2 = (0,1,0), P_3 = (0,0,1), P_4 = (1,1,0), P_5 = (1,0,1), P_6 = (0,1,1), P_7 = (1,1,1).$$

Its 7 lines are the triples of points whose (mod-2) sum is zero. The four lines of $\mathrm{PG}(2,2)$ not passing through P_1 are:

$$L_1 = \{P_2, P_3, P_6\}, L_2 = \{P_2, P_5, P_7\}, L_3 = \{P_3, P_4, P_7\}, L_4 = \{P_4, P_5, P_6\}.$$

Lemma 2.5 (Constructive S_4 embedding). *The point stabilizer $\mathrm{Stab}_{\mathrm{GL}(3,2)}(P_1)$ is isomorphic to S_4 and is generated over $\mathbb{Z}_2$ by the matrices*

$$g_1 = [[1,0,0],[0,0,1],[0,1,0]], \quad g_2 = [[1,1,0],[0,1,1],[0,0,1]]$$

satisfying the presentation $\langle g_1, g_2 \mid g_1^2 = g_2^4 = (g_1 g_2)^3 = I \rangle \cong S_4$.

Proof. We verify each claim by direct computation over $\mathbb{Z}_2$.

(1) *Membership in $\mathrm{GL}(3,2)$.* $\det(g_1) = 1(0 \cdot 0 - 1 \cdot 1) = 1 \pmod{2}$ and $\det(g_2) = 1(1 \cdot 1 - 1 \cdot 0) = 1$. Both matrices are invertible over $\mathbb{Z}_2$.

(2) *Stabilization of $P_1 = (1,0,0)^T$.* $g_1(1,0,0)^T = (1,0,0)^T = P_1$ and $g_2(1,0,0)^T = (1,0,0)^T = P_1$.

(3) *Orders.* $g_1^2 = I$ (verified by direct multiplication; g_1 is a reflection). $g_2^2 = [[1,0,1],[0,1,0],[0,0,1]] \neq I$, but $g_2^4 = I$, confirming g_2 has order 4.

(4) *Mixed relation.* One computes $g_1 g_2 = [[1,1,0],[0,0,1],[0,1,1]]$, and verifies $(g_1 g_2)^3 = I$ by direct matrix multiplication.

(5) *Faithful action on $\{L_1, L_2, L_3, L_4\}$.* Applying g_1 and g_2 to the four lines: $g_1 \mapsto (L_2 L_3)$, $g_2 \mapsto (L_1 L_4 L_2 L_3)$. These permutations have orders 2 and 4 respectively. Their product $(L_2 L_3)(L_1 L_4 L_2 L_3) = (L_1 L_4)$

L_3) has order 3, satisfying the presentation. The group $\langle g_1, g_2 \rangle$ therefore surjects onto S_4 . Since $|\text{Stab}(P_1)| = |\text{GL}(3,2)|/7 = 168/7 = 24 = |S_4|$, the surjection is an isomorphism. $\square$

2.4 Spectral Decomposition and Gap Stability

The coset space G/S_4 has 7 points, identified with the points of the Fano plane $PG(2,2)$.

Definition 2.6 (Coset Laplacian). Let $S = \{s \in G : s \text{ lies in class } 2A\}$, so $|S| = 21$. The coset Laplacian Δ is the operator on $L^2(G/S_4)$ defined by

$$(\Delta f)(P) = f(P) - (1/|S|) \sum_{s \in S} f(s \cdot P)$$

for $f : G/S_4 \rightarrow \mathbb{C}$ and $P \in G/S_4$.

Remark 2.7. Since G acts 2-transitively on the 7 points of $PG(2,2)$, each of the 21 class-2A involutions either fixes a coset or transposes two cosets. For a fixed coset P , the 9 involutions in S_4 fix P , and the remaining 12 move P to one of 6 other cosets (by 2-transitivity, exactly 2 involutions send P to each specific other coset Q). Thus $\sum_{s \in S} f(s \cdot P) = 9f(P) + 2 \sum_{Q \neq P} f(Q)$, and for $f \perp 1$:

$$(\Delta f)(P) = f(P) - (9f(P) - 2f(P))/21 = f(P) - 7f(P)/21 = (2/3)f(P).$$

Lemma 2.8 (Spectral decomposition). $L^2(PSL(2,7)/S_4) \cong \rho_1 \oplus \rho_6$. All other irreducible representations of $PSL(2,7)$ (of dimensions 3, 3, 7, 8) occur with multiplicity zero.

Proof. By Frobenius reciprocity [10, Theorem 7.2], the multiplicity of an irreducible ρ in $L^2(G/S_4) \cong \text{Ind}(S_4, G) 1$ equals $n_1(S_4; \rho)$. From the proof of Theorem 2.3: $n_1(S_4; \rho_6) = 1$. The trivial representation gives $n_1(S_4; \rho_1) = 1$ trivially. For the remaining four irreps, direct inner product calculations yield $n_1(S_4; \rho_3) = n_1(S_4; \rho_3') = n_1(S_4; \rho_7) = n_1(S_4; \rho_8) = 0$. The dimension check $1 + 6 = 7 = |G/S_4|$ confirms completeness. $\square$

Lemma 2.9 (Spectral gap and stability). The first non-trivial eigenvalue of Δ is $\lambda_1 = 2/3$. Moreover, every G -invariant operator on $L^2(G/S_4)$ acts as a scalar on the ρ_6 -isotypic component; the eigenvalue $2/3$ is an exact, not approximate, spectral identity for the specific operator Δ .

Proof. *Step 1 (Schur's lemma).* By Lemma 2.8, $L^2(G/S_4) \cong \rho_1 \oplus \rho_6$ with each irreducible appearing with multiplicity one. By Schur's lemma [10, Theorem 1.7], $\text{End}^G(L^2(G/S_4)) \cong \mathbb{C} \oplus \mathbb{C}$. Every G -equivariant operator on $L^2(G/S_4)$ acts as a scalar on each isotypic component.

Step 2 (G -invariance of Δ). The generating set S consists entirely of conjugacy class 2A, so S is closed under conjugation by G . Therefore Δ commutes with the left G -action: Δ is G -invariant. Any G -invariant perturbation $\Delta + V$ (where V commutes with the G -action) likewise decomposes into scalars on ρ_1 and ρ_6 .

Step 3 (Eigenvalue computation). All 21 elements of S lie in class 2A. By Schur's lemma, the operator $A = (1/|S|)\sum_{s \in S} \rho_6(s)$ equals a scalar $c \cdot I_6$ on ρ_6 . Taking traces: $c = \chi_{\rho_6}(2A)/\dim(\rho_6) = 2/6 = 1/3$. Therefore $\Delta = I - A$ acts on ρ_6 as $\lambda_1 = 1 - c = 2/3$.

Step 4 (Stability statement). The decomposition $\rho_1 \oplus \rho_6$ is determined by group theory alone and cannot be altered by G -invariant perturbations. A different G -invariant operator (with a different generating set) would preserve the decomposition but alter the scalar c ; the value $\lambda_1 = 2/3$ is the exact eigenvalue for the specific operator Δ defined above.

□

3. Cheeger Analysis

3.1 Structure of the Coset Graph

Definition 3.1 (Coset graph). The coset graph Γ has vertex set $V = G/S_4$, the 7 cosets of S_4 , identified with $\{P_1, \dots, P_7\}$. For each class-2A involution s and each $P \in V$, place a directed edge $(P, s \cdot P)$.

Proposition 3.2. *The underlying undirected coset graph Γ is the complete multigraph $2 \cdot K_7$: for every pair $\{P, Q\}$ with $P \neq Q$, the vertices P and Q are connected by exactly 2 edges. Each vertex has degree 12.*

Proof. Since G acts 2-transitively on the 7 points of $\text{PG}(2,2)$, the number of involutions sending a fixed P to a fixed $Q \neq P$ is constant over all ordered pairs (P, Q) . The total number of involution-vertex pairs (s, P) with $s \cdot P \neq P$ equals $|S| \cdot 7 - (\text{fixed pairs})$. The 21 involutions fix coset P_j if and only if $s \in S \cap S_4^{\text{gj}}$ (the conjugate of S_4 at P_j), which has size 9 (since S_4 contains exactly 9 involutions, all in class 2A). Thus each vertex has $21 - 9 = 12$ edges leading away from it, and the 12 edges from P distribute over 6 other vertices. By 2-transitivity, exactly $12/6 = 2$ edges connect P to each other vertex Q .

□

3.2 Cheeger Constant

Definition (Random-walk Cheeger constant). For a subset $A \subseteq V$ with $|A| = k \leq \lfloor n/2 \rfloor$, the expansion ratio of A is $h(A) = |\{(s, P) : s \in S, P \in A, s \cdot P \notin A\}| / (|S| \cdot |A|)$. The Cheeger constant is $h = \min$ over all such A of $h(A)$.

Theorem 3.3 (Cheeger constant). *The Cheeger constant of Γ is $h = 8/21$, attained at any cut of size $k = 3$. The discrete Cheeger inequality $h^2/2 \leq \lambda_1 \leq 2h$ holds. Neither bound is an equality.*

Proof. *Computing h .* For a cut of size k , the expansion ratio is $h_k = 2k(7-k)/(|S| \cdot k) = 2(7-k)/21$. Tabulating for $k = 1, 2, 3$:

k	exit pairs	h_k
1	12	12/21
2	20	10/21

3	24	8/21
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The minimum is $h = 8/21$, attained at $k = 3$.

Verifying the Cheeger inequality. Lower bound: $h^2/2 = (8/21)^2/2 = 64/882 = 32/441 \approx 0.073 \leq 2/3 = \lambda_1$.

✓

Upper bound: $\lambda_1 = 2/3 = 14/21 \leq 16/21 = 2 \cdot 8/21 = 2h$. ✓

Non-sharpness. Both inequalities are strict: $32/441 < 2/3$ and $14/21 < 16/21$.

□

4. Discussion

4.1 Three Distinct Spectral Quantities

Table 2 summarizes three spectral quantities that arise naturally in the geometry of $G = PSL(2,7)$ and that must be kept distinct.

Table 2: Spectral quantities

Quantity	Value	Domain	Meaning
λ_1	$2/3$	$L^2(G/S_4)$, 7-vertex coset graph	Eigenvalue of the coset Laplacian Δ
δ	$10/21$	$L^2(G)$, full group spectrum	Gap between ρ_6 and ρ_7 eigenvalues
h	$8/21$	Coset graph $2 \cdot K_7$	Random-walk Cheeger constant

The quantity $\lambda_1 = 2/3$ is a property of the operator Δ on the 7-dimensional coset space, computed in Lemma 2.9. The inter-representation gap $\delta = 10/21$ is the difference between the eigenvalues of Δ on ρ_7 and on ρ_6 in the full group Laplacian Δ^G on $L^2(G)$; it is an algebraic property of the representation theory of $PSL(2,7)$ that does not directly involve the coset space G/S_4 . The Cheeger constant $h = 8/21$ is an isoperimetric property of the graph Γ , computed in Theorem 3.3. These three quantities are logically independent.

Open Problem 4.1. The Cheeger ratio at 2-vertex cuts is $h_2 = 10/21 = \delta$. No structural explanation for this coincidence is known. In particular, there is no established map between the isoperimetric problem on Γ and the representation-theoretic gap in $L^2(G)$. With the generating set S , the eigenvalues of the full group Laplacian on $L^2(G)$ are 0, $2/3$, 1, $8/7$ and $4/3$ on ρ_1 , ρ_6 , ρ_8 , ρ_7 and ρ_3 , so δ is one of several differences among them (the next eigenvalue above $2/3$ belongs to ρ_8); the coincidence should be weighed against that choice.

4.2 The Smith Obstruction

We record a structural constraint on the global topology of orbifolds associated to G , independent of the spectral results above.

Proposition 4.2 (Smith obstruction). *The group $G = PSL(2,7)$ cannot act freely on any sphere S_{n-1}^n for any n . Consequently, S^7/G is an orbifold, not a manifold.*

Proof. The Sylow 2-subgroup of G is isomorphic to D_4 (it appears as the flag stabilizer in Appendix A). This subgroup D_4 contains the Klein four-group $V_4 = \mathbb{Z}_2 \times \mathbb{Z}_2$, specifically the subgroup generated by any two commuting involutions. The four-group V_4 is a non-cyclic abelian 2-group. By Smith's theorem [11]: if a finite group G contains a non-cyclic abelian p -subgroup, then G cannot act freely on any sphere S^n . Since $V_4 \subset G$ is a non-cyclic abelian 2-subgroup, the conclusion follows. $\square$

Remark 4.3. The Smith obstruction is independent of the spectral gap stability established in Lemma 2.9. The gap stability is a finite-dimensional algebraic fact following from Schur's lemma applied to the coset Laplacian; it requires no topology. The Smith obstruction is a statement about continuous actions on spheres [2]; it does not bear on the eigenvalue $\lambda_1 = 2/3$. These are parallel results about the same group, not causally related.

4.3 Open Problems

Open Problem 4.4 (Orbifold index theory for $S^7/\mathrm{PSL}(2,7)$). By Proposition 4.2, the quotient S^7/G , for the action of G on $S^7 \subset \mathbb{R}^8$ through its 8-dimensional irreducible representation ρ_{81} , is a 7-dimensional orbifold with non-isolated singular strata: S^3 strata from the 21 class-2A elements (codimension 4) and S^1 strata from classes 3A, 4A, 7A, 7B (codimension 6, total 104146 elements; the circle fixed by an element of order 4 lies inside the 3-sphere fixed by its square). The Kawasaki orbifold index theorem [8] applies only to isolated singularities. The mixed-dimensional, non-isolated stratification present in S^7/G requires either the groupoid index theory of Pflaum–Posthuma–Tang [9] or the L^2 -cohomology methods for singular spaces (Cheeger). The η -invariant of $S^7/\mathrm{PSL}(2,7)$ is currently unknown. Because S^7/G is the quotient of a closed manifold by a finite group, index problems on it reduce to G -equivariant ones on S^7 : the index on G -invariant sections is the average over G of the Lefschetz numbers given by the Atiyah–Singer fixed-point formula, whose fixed-point sets may have any dimension [13]. Kawasaki's index theorem for compact V -manifolds [17], whose complex case is the Riemann–Roch theorem of [8], treats the same problem intrinsically, and groupoid methods [9] extend it further. The η -invariant of $S^7/\mathrm{PSL}(2,7)$ is correspondingly the average over G of equivariant η -invariants [14], a finite sum over conjugacy classes computable from the fixed-point data above; it is not computed here.

Open Problem 4.5 (Uniform flag stabilizer description). The three maximal subgroup classes of $\mathrm{PSL}(2,7) \cong \mathrm{GL}(3,2)$ admit natural incidence-geometric descriptions in $\mathrm{PG}(2,2)$. The two S_4 classes are the point-stabilizer $S_4 = \mathrm{Stab}(P_1)$ (Lemma 2.5) and the line-stabilizer $S_4 = \mathrm{Stab}(\ell)$, interchanged by the point-line duality. The Borel subgroup $Z_7 \rtimes Z_3$ is the normalizer of a Singer cycle, fixing no point or line; a constructive $\mathrm{GL}(3,2)$ presentation of it is not given in this work. The flag stabilizer $\mathrm{Stab}(P_1, L_{124}) \cong D_4$ (Appendix A) is the Sylow 2-subgroup and is not maximal. A uniform framework relating these stabilizers within the incidence geometry of $\mathrm{PG}(2,2)$ — and describing how the point-line duality permutes them — would be natural.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

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Appendix A. Constructive $\mathrm{GL}(3,2)$ Generators for D_4 as Flag Stabilizer

We realize $M_3 \cong D_4$ as the stabilizer in $\mathrm{GL}(3,2)$ of the flag (P_1, L_{124}) , where $P_1 = (1,0,0)$ and $L_{124} = \mathrm{span}_{\mathbb{Z}_2}\{e_1, e_2\} = \{P_1, P_2, P_4\}$ is the projective line through P_1 and P_2 .

Lemma A.1 (D_4 flag stabilizer). *The flag stabilizer $\mathrm{Stab}_{\mathrm{GL}(3,2)}(P_1, L_{124})$ is isomorphic to D_4 and is generated over $\mathbb{Z}_2$ by*

$$h_1 = [[1,1,0],[0,1,0],[0,0,1]], \quad h_2 = [[1,0,0],[0,1,1],[0,0,1]]$$

satisfying the presentation $\langle h_1, h_2 \mid h_1^2 = h_2^2 = (h_1 h_2)^4 = I \rangle \cong D_4$.

Proof. We verify each claim by direct computation over $\mathbb{Z}_2$.

(1) *Membership.* $\det(h_1) = 1$ and $\det(h_2) = 1$ over $\mathbb{Z}_2$; both are in $\mathrm{GL}(3,2)$.

(2) *Stabilization of P_1 .* $h_1(1,0,0)^T = (1,0,0)^T = P_1$; $h_2(1,0,0)^T = (1,0,0)^T = P_1$.

(3) *Preservation of L_{124} .* The line $L_{124} = \{P_1, P_2, P_4\}$ corresponds to the projective points of $\mathrm{span}(e_1, e_2) = \{0, e_1, e_2, e_1+e_2\}$. The matrix h_1 acts as $(x,y,z) \mapsto (x+y, y, z)$: $h_1(P_1) = P_1$, $h_1(P_2) = P_4$, $h_1(P_4) = P_2$. The matrix h_2 acts as $(x,y,z) \mapsto (x, y+z, z)$: $h_2(P_1) = P_1$, $h_2(P_2) = P_2$, $h_2(P_4) = P_4$. Both generators fix P_1 and map $\{P_1, P_2, P_4\}$ into itself. ✓

(4) *Orders.* $h_1^2(x,y,z) = h_1(x+y, y, z) = (x+y+y, y, z) = (x,y,z)$, so $h_1^2 = I$. $h_2^2(x,y,z) = h_2(x, y+z, z) = (x, (y+z)+z, z) = (x,y,z)$, so $h_2^2 = I$.

(5) *Product order.* $h_1 h_2 = [[1,1,1],[0,1,1],[0,0,1]]$. Let $A = h_1 h_2$. Direct computation gives $A^2 = [[1,0,1],[0,1,0],[0,0,1]] \neq I$ and $A^4 = (A^2)^2 = I$. So $h_1 h_2$ has order 4.

(6) *Non-abelian.* $h_2 h_1 = [[1,1,0],[0,1,1],[0,0,1]] \neq h_1 h_2$.

(7) *Order of the group.* From the presentation $h_1^2 = h_2^2 = (h_1 h_2)^4 = I$ with $h_1 h_2 \neq h_2 h_1$, the group $\langle h_1, h_2 \rangle$ is a non-abelian group of order 8; enumerating $\{I, h_1, h_2, h_1 h_2, (h_1 h_2)^2, h_1(h_1 h_2)^2, (h_1 h_2)^3, h_1(h_1 h_2)^3\}$ shows $|\langle h_1, h_2 \rangle| = 8$. Since Q_8 has a unique element of order 2 but h_1 and h_2 are two distinct involutions, $\langle h_1, h_2 \rangle \cong D_4$. ✓

The index $|G|/|\langle h_1, h_2 \rangle| = 168/8 = 21$ equals the number of flags (incident point-line pairs) in $\mathrm{PG}(2,2)$, confirming that $\langle h_1, h_2 \rangle$ is the full flag stabilizer. It is a Sylow 2-subgroup of G , contained in S_4 and therefore not maximal (cf. Remark 2.2).

□